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Achieving Sub-7-Minute Epochs: A Blueprint for Optimizing Vesuvius Challenge Training on H100

This report provides a comprehensive analysis and actionable recommendations for optimizing the training process of a deep learning model for the Vesuvius Challenge. The primary objective is to reduce the epoch duration from 30 minutes to the target range of 6-7 minutes by leveraging a powerful H100 GPU with 80GB of VRAM and 220GB of system RAM. The analysis synthesizes findings from extensive troubleshooting, log file examination, and code modification attempts to address critical bottlenecks in data loading, computational throughput, and numerical stability. The proposed strategies focus on maximizing hardware utilization while ensuring robust and effective model learning.

Diagnosing the Primary Bottlenecks in the Training Pipeline

The initial problem of a 30-minute epoch time indicates significant underutilization of the H100 GPU. Analysis of the provided materials reveals several interconnected bottlenecks that contribute to this inefficiency. The first identified bottleneck was likely GPU underutilization due to a slow data pipeline. This led to the exploration of standard optimization levers such as increasing the batch size (`BATCH_SIZE`), the number of DataLoader workers (`num_workers`), and prefetching batches ahead of time [[user's_4th_response]]. While these adjustments are fundamental, they may not be sufficient if the core operations within the training loop remain a limiting factor. The second, and more critical, bottleneck was traced to the CPU-bound nature of the "heavy" data augmentation pipeline. Even with preloaded data, complex augmentations using libraries like `scipy.ndimage` were consuming significant CPU cycles, forcing the GPU to wait for processed batches, thus creating idle time [[user's_6th_response]]. The third major issue was numerical instability, manifesting as NaN gradients during training, especially when using Automatic Mixed Precision (AMP). This instability, likely originating from the interaction between AMP and complex topology-aware loss

functions, caused batches to be skipped, disrupting the training flow and hindering model learning [[user's_15th_response]]. Finally, the learning rate (LR) scheduling mechanism proved to be ineffective; after resuming training, the LR decayed extremely slowly, preventing the optimizer from making finer adjustments necessary for convergence in later epochs [[user's_22nd_response]]. These issues demonstrate that achieving high-speed training requires a holistic approach, addressing not just data movement but also computational efficiency, numerical precision, and optimization dynamics.

A key finding from the investigation was that the root cause of the slow training was not just a lack of parallelism but also inefficient computational kernels. The original notebook used `scipy.ndimage` for augmentations, which, despite being efficient for its purpose, still represents a CPU-bound workload that cannot saturate the H100's processing units. Furthermore, the reliance on AMP introduced numerical instabilities that compromised training stability. The solution path involved moving away from CPU-intensive augmentations towards GPU-accelerated alternatives or highly optimized CPU logic, while simultaneously stabilizing the training loop by addressing the sources of NaNs. The table below summarizes the identified bottlenecks and their direct impact on training speed.

Bottleneck Category	Specific Issue Identified	Impact on Training
Data Pipeline	CPU-bound augmentations using `scipy.ndimage` [[user's_6th_response]]	GPU remains idle waiting for augmented data, leading to low overall throughput.
Computational Efficiency	Use of standard `nn.InstanceNorm3d` with AMP	Leads to numerical overflow/underflow in FP16, causing NaN gradients and unstable training [[user's_18th_response]].
Optimization Dynamics	Slow LR decay after resuming training [[user's_22nd_response]]	Model struggles to converge, taking longer to find optimal weights and potentially getting stuck in local minima.
Model Architecture	Complexity of topology-aware loss functions (e.g., `MedialSurfaceRecall`)	These losses introduce non-differentiable or numerically sensitive operations that exacerbate instability under AMP [[user's_15th_response]].

Addressing these bottlenecks requires a multi-pronged strategy. For the data pipeline, leveraging the ample system RAM (220 GB) to preload all data into memory is the first step to eliminate disk I/O latency. The next step is to ensure the augmentation logic itself is as fast as possible. While `scipy.ndimage` is generally fast, the true path to maximum throughput lies in offloading these computations to the GPU using libraries like MONAI, which can accelerate operations via cuCIM. However, given the power of modern CPUs and the overhead of transferring small patches to the GPU, highly optimized CPU-based augmentations often provide the best balance of speed and simplicity. For computational efficiency, replacing `nn.InstanceNorm3d` with a numerically stable variant like `SafeInstanceNorm3d` is a prerequisite for successfully using AMP

[[user's_18th_response]]. This allows the use of FP16 precision, which the H100's tensor cores are designed to accelerate. Finally, the learning rate schedule must be carefully managed. After resuming training, the scheduler should be reset or restarted to ensure the LR decays at an appropriate rate for the remaining epochs, rather than continuing a long, ineffective decay schedule.

Strategies for Accelerating Data Loading and Augmentation

To significantly reduce epoch times, the data loading and augmentation pipeline must be optimized for maximum throughput. Given the availability of 220 GB of system RAM, preloading the entire dataset into memory is not only feasible but is a critical first step to eliminate disk I/O bottlenecks. The `VesuviusDatasetV3` class in the provided notebooks already implements this strategy, which ensures that subsequent access to volume data is a near-instantaneous memory read rather than a slower disk seek operation [[user's_4th_response]]. This foundational step transforms the challenge from one of raw data retrieval speed to one of computational speed for augmentation and model inference.

With data preloaded, the focus shifts to the efficiency of the augmentation logic itself. The `vesuvius_v6_training.ipynb` notebook demonstrates a comprehensive set of nnU-Net-style augmentations, including spatial transformations (rotation, scaling), intensity variations (noise, blur, brightness, contrast, gamma), and geometric flips (mirror) [[user's_6th_response]]. These augmentations are crucial for generalization, and simply removing them would compromise model quality. The goal is to execute them as quickly as possible. The `augment_spatial_fast` function in the modified notebook uses `scipy.ndimage.zoom` and `ndimage.affine_transform` with minimal overhead, which is a strong baseline for CPU-based speed. To further accelerate this, several strategies can be considered:

1. **Leverage GPU-Accelerated Augmentations:** If profiling shows the CPU is still a bottleneck, offloading augmentations to the GPU is the most powerful option. Libraries like MONAI provide GPU-accelerated versions of common transforms (e.g., `Rand3DElasticd`). By moving augmentations to the GPU, the data never leaves the GPU's high-bandwidth memory, eliminating the CPU-to-GPU transfer bottleneck. This requires careful implementation, as some complex transforms might need to be adapted or replaced, but for many standard operations (like

rotation, scaling, and simple blurs), GPU acceleration can yield a substantial speedup. The trade-off is increased complexity and potential memory usage on the GPU.

2. **Vectorize Operations:** Wherever possible, replace loops over individual patches with vectorized NumPy operations. NumPy is highly optimized and can perform element-wise operations on entire arrays much faster than Python loops. While the patch-based nature of the dataset makes this difficult, ensuring that augmentations operate on the whole volume slice at once before cropping a patch can help.
3. **Reduce Augmentation Complexity:** The nnU-Net pipeline is intentionally complex to maximize diversity. However, for the sake of speed, one could experiment with simplifying it. For instance, reducing the probability of applying each augmentation (`AUG_ROTATION_P`, `AUG_GAUSSIAN_NOISE_P`, etc.) or shortening the range of values (e.g., `AUG_SCALE_RANGE`, `AUG_GAMMA_RANGE`) can decrease the computational load per batch. This is a trade-off between training speed and the richness of the data distribution the model sees.
4. **Tune System-Level Parameters:** The PyTorch `DataLoader` offers parameters that control its parallelism. Increasing `num_workers` allows more CPU cores to be dedicated to data loading and preprocessing. With an H100 and a powerful CPU, setting `num_workers` to a higher value (e.g., 16 or 20) can help keep up with the GPU's demands. Similarly, increasing `prefetch_factor` ensures that multiple batches are always queued and ready, hiding any remaining latency from the GPU. The combination of these settings creates a highly parallel pipeline where data preparation happens concurrently with model computation.

By combining data preloading with a finely-tuned augmentation pipeline and optimized system parameters, the data bottleneck can be largely eliminated, allowing the GPU to operate at peak efficiency. The choice between a highly optimized CPU pipeline and a GPU-accelerated one depends on the specific profile of the augmentations and the observed bottleneck during training.

Maximizing GPU Utilization through Batch Configuration and AMP

Achieving the sub-7-minute epoch goal hinges on maximizing the utilization of the H100 GPU's immense computational power. The most direct way to do this is by aggressively increasing the `BATCH_SIZE`. The epoch time of approximately 400 seconds (~6-7 minutes) observed in the logs corresponds to a `BATCH_SIZE` of 16

[[user's_22nd_response]]. The H100's 80GB of VRAM can comfortably accommodate larger batches, especially with smaller patch sizes like `(128, 128, 128)`. Pushing the batch size to `32`, `64`, or even higher will increase the amount of work done per forward/backward pass, thereby reducing the number of steps required per epoch and directly cutting down wall-clock time. For example, doubling the batch size from 16 to 32 would halve the number of iterations per epoch, assuming memory permits.

However, increasing the batch size necessitates a corresponding increase in the learning rate to maintain similar optimization dynamics. A common heuristic is the square root rule: $new_lr = old_lr \times \sqrt{new_bs / old_bs}$. For instance, increasing the batch size from 16 to 64 (a 4x increase) would require multiplying the learning rate by $\sqrt{4} = 2$. This ensures the magnitude of parameter updates remains consistent. The `Config` class in the training notebook is the place to adjust both `BATCH_SIZE` and `LEARNING_RATE`.

Another critical component for H100 performance is the correct use of Automatic Mixed Precision (AMP). The H100's tensor cores provide a significant performance boost for `float16` (FP16) arithmetic compared to the default `float32` (FP32). Enabling `USE_AMP=True` in the configuration can dramatically accelerate both forward and backward passes. However, as previously established, AMP can introduce numerical instabilities, leading to NaN gradients. Therefore, enabling AMP is conditional on ensuring numerical stability throughout the model and loss functions. This involves using a stable normalization layer like `SafeInstanceNorm3d` and potentially disabling or modifying problematic components of the loss function [[user's_18th_response]]. Once stability is confirmed, AMP becomes a mandatory optimization for reaching the target training speed.

Finally, the model itself should be compiled for the target hardware. Using `torch.compile()` on the model before training can optimize the execution graph, fusing operations and generating more efficient machine code tailored to the H100's architecture. This can provide additional speedups beyond those gained from AMP alone. The combination of a large batch size, a properly scaled learning rate, enabled AMP, and model compilation forms the cornerstone of a high-throughput training strategy on the H100. The table below outlines the recommended changes to the configuration to maximize GPU utilization.

Parameter	Default Value	Recommended Value	Rationale
`BATCH_SIZE`	2	32 or 64	Increases data processed per step, saturating GPU compute units. Requires memory check.
`LEARNING_RATE`	0.01	~0.028 (for bs=32)	Scales the learning rate proportionally to the batch size increase.
`USE_AMP`	False	True	Enables FP16 training, leveraging H100's tensor cores for significant speedup.
`NUM_WORKERS`	8	16 or 20	Increases CPU parallelism for data loading and augmentation to feed the GPU.
`PREFETCH_FACTOR`	4	8	Ensures a constant queue of batches is available, hiding data loading latency.

Implementing these changes systematically, starting with increasing the batch size and learning rate, followed by enabling AMP, will unlock the full potential of the H100 hardware and bring the epoch time well within the desired 6-7 minute window.

Ensuring Numerical Stability and Effective Learning Dynamics

While pushing for speed is paramount, it is equally important to ensure the training process is stable and conducive to learning. The experience with NaN gradients serves as a critical lesson that aggressive optimization can sometimes destabilize the model. The primary source of these instabilities was traced back to the interaction between Automatic Mixed Precision (AMP) and the complex topology-aware loss functions (`MedialSurfaceRecall`, `SoftClDiceLoss`) [[user's_15th_response]]. These losses involve intricate mathematical operations that can become numerically unstable when performed in lower-precision `float16` format, leading to gradients that become `NaN`. To mitigate this, a two-pronged strategy is recommended.

First, the training loop itself must incorporate robust checks for numerical instability. The `train_one_epoch` function in the `vesuvius_v3_safe_amp_training.ipynb` notebook correctly implements this by checking for `torch.isnan(losses['total'])` after the forward pass and, if found, skipping the batch update. It also checks for `torch.isnan(param.grad).any()` before the optimizer step. These checks prevent the model from diverging catastrophically due to a single bad batch and allow training to continue, albeit with slightly less data per epoch

[[user's_18th_response]]. Implementing this logic is essential for any training run involving AMP.

Second, a more proactive approach is needed to handle the unstable components of the loss function. The evidence strongly suggests that the topology losses are the culprits. Therefore, a practical and effective strategy is to adopt a **two-phase training schedule**:
1. **Phase 1 (Stable Pre-training):** For the majority of the training run (e.g., epochs 0 to 800), configure the loss function to use only the stable Dice and Cross-Entropy components. Set `SKELETON_WEIGHT=0.0` and `CLDICE_WEIGHT=0.0`. During this phase, the model focuses on learning the base segmentation task with a reliable loss signal, free from the disruptive influence of the unstable topology losses. 2. **Phase 2 (Topological Fine-Tuning):** In the final stages of training (e.g., epochs 801 to 1200), gradually reintroduce the topology-aware losses with reduced weights (e.g., `SKELETON_WEIGHT=0.1`, `CLDICE_WEIGHT=0.05`). By this point, the model's weights are well-initialized and the loss landscape is likely smoother, making it more resilient to the minor instabilities of the topology losses. This fine-tuning phase aims to nudge the model's predictions towards better topological properties without derailing the entire training process.

This phased approach isolates the instability, allowing the model to benefit from the regularization effects of the topology losses without being derailed by them. It represents a pragmatic compromise between achieving maximum training stability and attempting to optimize for the Topology Score component of the leaderboard metric.

Furthermore, managing the learning rate schedule effectively is crucial for guiding the model to a good minimum. As noted, the default scheduler can lead to overly slow decay. A more adaptive strategy, such as `torch.optim.lr_scheduler.ReduceLROnPlateau`, can be employed. This scheduler monitors a metric (like validation loss) and only reduces the LR when the metric has stopped improving for a certain number of epochs (`patience`). This prevents premature reduction of the LR and allows the model to explore the loss landscape more thoroughly before settling into smaller steps. Alternatively, a `StepLR` scheduler, which halves the LR every fixed number of epochs, can provide more predictable and aggressive decay. Integrating such a scheduler ensures that the optimization process remains dynamic throughout the entire training run.

An Integrated Action Plan for Sub-7-Minute Epochs

To successfully transition from a 30-minute epoch to the target of 6-7 minutes, a holistic and integrated strategy is required. This plan synthesizes the key insights from the analysis of bottlenecks, data pipelines, computational efficiency, and training stability. It provides a concrete, step-by-step roadmap for implementing the necessary changes to the training environment and codebase.

Step 1: Refactor the Configuration and Enable High-Throughput Settings

The foundation of the optimization lies in the configuration. Create a new configuration file or modify the existing `Config` class to reflect the following changes, which are designed to maximize hardware utilization.

Parameter	Change From	Change To	Rationale
`BATCH_SIZE`	2 or 8	32	Drastically increases data processed per GPU step, directly reducing epoch time. Check VRAM limits.
`LEARNING_RATE`	0.028	0.053	Scaled using the square root rule ($0.028 * \sqrt{32/8}$) to match the new batch size.
`USE_AMP`	True	True	Essential for leveraging H100's tensor cores. Ensure `SafeInstanceNorm3d` is implemented.
`NUM_WORKERS`	12	16	Increases CPU parallelism for data loading and augmentation. Adjust based on CPU core count.
`PREFETCH_FACTOR`	8	8	Maintains a large buffer of batches to hide any remaining data loading latency.
`PATCH_SIZE`	(128, 128, 128)	(128, 128, 128)	Keep the patch size to maintain a good balance between context and memory usage for the larger batch size.
`RUN_VAL_SWEEP`	True	False	Disables the local validation sweep that incorrectly accesses training data paths.

Step 2: Implement a Two-Phase Training Schedule

To manage the instability of topology losses while pursuing high performance, implement a learning rate scheduler that enables a two-phase training approach. Modify the `main_training` function to include logic that adjusts the loss weights after a certain number of epochs.

```
# Inside main_training() after defining the scheduler
phase_1_epochs = 800 # Number of epochs for stable pre-training

for epoch in range(start_epoch, cfg.EPOCHS):
    # --- Phase Logic ---
```

```
if epoch < phase_1_epochs:
    # Phase 1: Stable training with only Dice and CE loss
    criterion.skeleton_weight = 0.0
    criterion.cldice_weight = 0.0
else:
    # Phase 2: Fine-tuning with topology losses
    criterion.skeleton_weight = cfg.SKELETON_WEIGHT
    criterion.cldice_weight = cfg.CLDICE_WEIGHT

# ... rest of the training loop ...
scheduler.step()
```

This structure ensures the model learns a robust base representation before being exposed to the complexities of topological optimization, mitigating the risk of NaN gradients.

Step 3: Optimize the Data Augmentation Pipeline

Ensure the augmentation functions are as efficient as possible. The `apply_nnunet_augmentations` function from the `vesuvius_v6_training.ipynb` notebook is a strong candidate. Profile the training loop to confirm that the CPU is not becoming a bottleneck. If the CPU is saturated and the GPU is still underutilized, consider selectively moving augmentations to the GPU using MONAI. For example, spatial augmentations like elastic deformations (`Rand3DElasticd`) can be GPU-accelerated.

Step 4: Ensure Numerical Stability

Before starting the training run, verify that the `SafeInstanceNorm3d` class is defined and used in the model architecture. This is a non-negotiable requirement for stable AMP-based training. Additionally, ensure the `train_one_epoch` function contains robust checks for `NaN` losses and gradients, as demonstrated in the provided safe training notebooks. This acts as a safety net against unexpected numerical issues.

Step 5: Execute and Monitor

Execute the training job with the newly configured settings. Monitor the training logs closely, paying attention to the following:

- **Epoch Time:** Confirm that the time per epoch converges to the 6-7 minute target.

- **GPU Utilization:** Use `watch -n 1 nvidia-smi` to monitor GPU utilization. It should be consistently high (above 80%).
- **NaN Batches:** Watch for the appearance of `>>> NaN gradient at batch...! Skipping...` messages. If they appear, it indicates the two-phase strategy may need adjustment, or AMP should be disabled.
- **Validation Score:** Track the validation Dice score to ensure the model is learning effectively.

By systematically implementing this integrated plan, you can transform the training pipeline from a slow, unstable process into a highly efficient, high-throughput engine. This will not only meet the strict timing requirements of the competition but also result in a more robust and better-performing model.